

Lecture 7 - Part IV

- Choice from Non-Linear Budget Sets
- Revealed Preferences
- Marginal Utility

ECON 101
Fall 2020



Suppose that $I = 100$, $P_F = 1$, $P_G = 1$

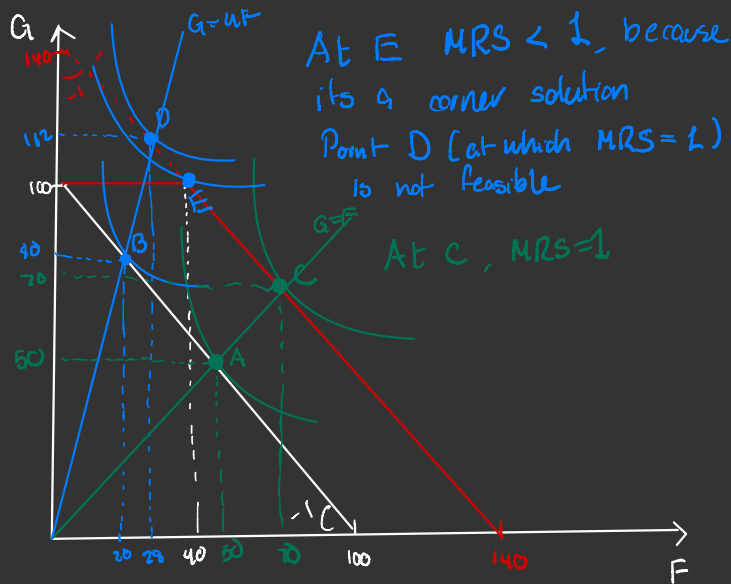
- Original Budget Line: $F + G = 100$

↳ eq.: $G = 100 - F$

- Receive \$40 worth of Food Stamps

$$G = \begin{cases} 100 & 0 \leq F \leq 40 \\ 140 - F & 40 < F \leq 140 \end{cases}$$

The dotted line + full line is $140 = F + G$



Consider two consumers:

$$U(F, G) = FG$$

$$MRS = \frac{G}{F}$$

$$U(F, G) = FG^4$$

$$MRS = \frac{G}{4F}$$

Before: $F + G = 100$

$$\frac{G}{F} = 1 \Rightarrow G = F$$

Plug this into B.L.

$$F + F = 100 \Rightarrow F = 50 = G$$

$$A = (50, 50)$$

$$U(50, 50) = 50 \times 50 = 2500$$

$$\frac{G}{4F} = 1 \Rightarrow G = 4F$$

Plug this into B.L.

$$F + 4F = 100 \Rightarrow 5F = 100$$

$$F = 20, G = 80, B = (20, 80)$$

$$U(20, 80) = 20 \times 80^4$$

After food stamps: use auxiliary B.L. $F + G = 140$

$$\frac{G}{F} = 1 \Rightarrow G = F$$

Plug this into B.L.

$$F + F = 140 \Rightarrow F = 70 = G$$

$$C = (70, 70)$$

$$U(70, 70) = 70^2$$

$$U(40, 100) = 40 \times 100^4$$

$$\frac{G}{4F} = 1 \Rightarrow G = 4F$$

Plug this into B.L.

$$F + 4F = 5F = 140 \Rightarrow F = \frac{140}{5} = 28$$

$$G = 4(28) = 112$$

$$D = (28, 112), U(28, 112) = 28 \times 112^4$$

$$E = (40, 100)$$

$$MRS = \frac{100}{4 \times 40} = \frac{5}{8} < 1 = \frac{P_F}{P_G}$$

Example: Quantity Discounts:

Suppose there are two goods x_1, x_2

Suppose $p_2 = 1$, $I = 100$

Quantity discount for x_1 :

$$-p_1 = 2 \text{ for } 0 \leq x_1 \leq 20 \Rightarrow \frac{p_1}{p_2} = 2$$

$$-p_1 = 1 \text{ for } x_1 > 20 \Rightarrow \frac{p_1}{p_2} = 1$$

This B.L. has two parts:

$$\begin{aligned} 0 \leq x_1 \leq 20 : & \quad 2x_1 + x_2 = 100 \Rightarrow x_2 = 100 - 2x_1 \\ x_1 > 20 : & \quad x_1 + x_2 = 80 \Rightarrow x_2 = 80 - x_1 \end{aligned}$$

Green budget line needs to pass through $(20, 60)$

$$\tilde{I} = (1)(20) + (1)(60) = 80$$

OR
by slope-point formula: $x_2 - 60 = -1(x_1 - 20) \Rightarrow x_2 = 80 - x_1$

$$U(x_1, x_2) = x_1 x_2, \quad MRS = \frac{x_2}{x_1}$$

To solve the problem, compute the solution for the two B.L.'s that constitute the combined one, and see which solution is consistent.

Combined B.L. with quantity discounts

$$x_2 = \begin{cases} 100 - 2x_1 & \text{if } 0 \leq x_1 \leq 20 \\ 80 - x_1 & \text{if } x_1 > 20 \end{cases}$$

$$\text{For } - : \frac{x_2}{x_1} = 2 \Rightarrow x_2 = 2x_1$$

$$\rightarrow 2x_1 + 2x_1 = 100 \Rightarrow x_1 = \frac{100}{4} = 25$$

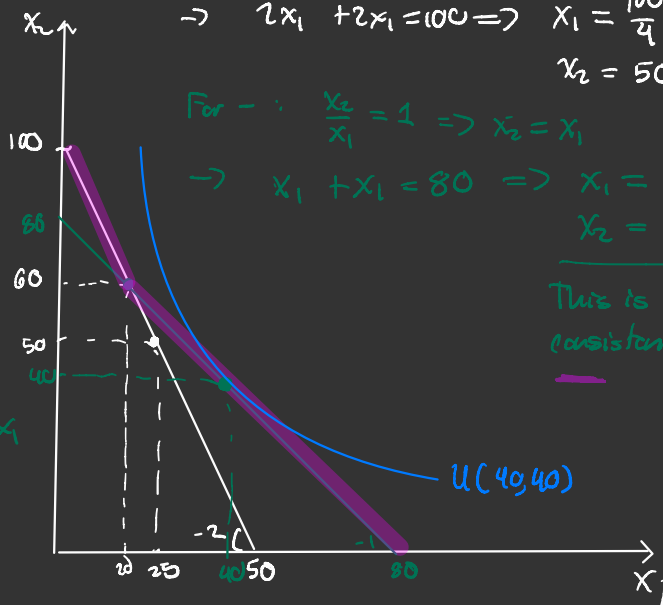
$$x_2 = 50$$

$$\text{For } - : \frac{x_2}{x_1} = 1 \Rightarrow x_2 = x_1$$

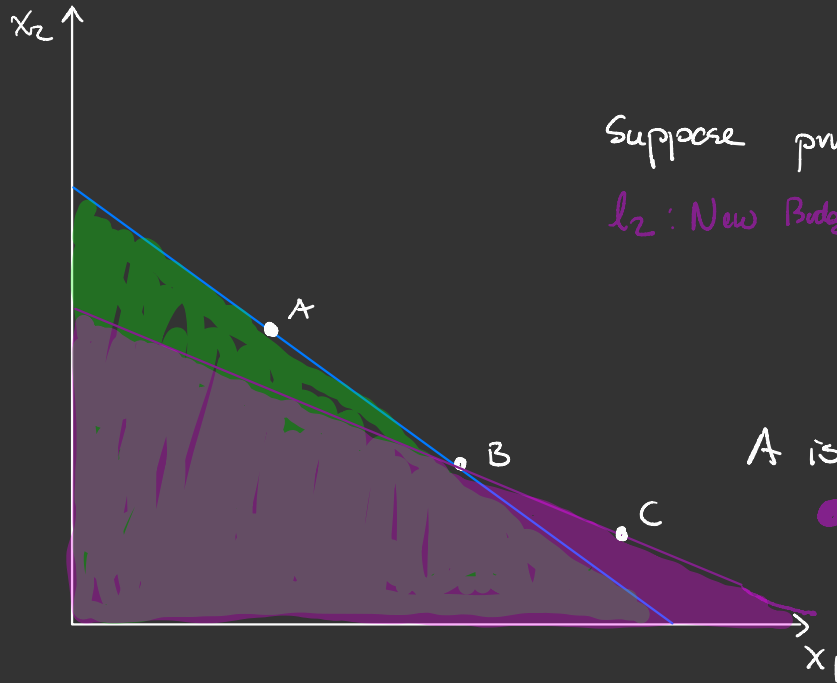
$$\rightarrow x_1 + x_1 = 80 \Rightarrow x_1 = 40$$

$$x_2 = 40$$

This is the sol.
consistent with



Revealed Preference



l_1 : B. constraint consumer chooses A, A is revealed preferred to everything in $\bullet$, in particular, point B

Suppose prices change, new budget line is

l_2 : New Budget. constraint, consumer chooses B, is revealed preferred to everything in $\bullet$, in particular point C.

A is revealed preferred to everything in $\bullet + \bullet$

Marginal Utility

At an indifference curve, $\bar{U} = U(x_1, x_2)$

Let's compute the change in utility at an IC when we change the consumption bundle:

$\Delta \bar{U} = 0$, because by definition, utility doesn't change at an I.C.

$$0 = MU_{x_1} \Delta x_1 + MU_{x_2} \Delta x_2 \Rightarrow$$

$$\text{rate of change} := \frac{\Delta x_2}{\Delta x_1} = - \frac{MU_{x_1}}{MU_{x_2}}$$

across IC



This is the slope of the IC, which is the $-MRS$

$$MRS = \frac{MU_{x_1}}{MU_{x_2}}$$

Example: perfect substitutes

$$U(x_1, x_2) = ax_1 + bx_2$$

$$\left. \begin{array}{l} MU_{x_1} = a \\ MU_{x_2} = b \end{array} \right\} MRS = \frac{a}{b} = \frac{MU_{x_1}}{MU_{x_2}}$$

$$U(x_1, x_2) = x_1 x_2 \quad \left\{ \begin{array}{l} MU_{x_1} = x_2 \\ MU_{x_2} = x_1 \end{array} \right.$$

$$MRS = \frac{x_2}{x_1} = \frac{MU_{x_1}}{MU_{x_2}}$$

Note: Marginal utilities are partial derivatives of the utility function.
Not required for this class

The concept of marginal utilities allows us to reinterpret the optimality condition

$$MRS = \frac{P_1}{P_2}$$

$$\Rightarrow \frac{MU_1}{MU_2} = \frac{P_1}{P_2} \Rightarrow \frac{MU_1}{P_1} = \frac{MU_2}{P_2} \quad \text{equimarginality principle}$$

"marginal utility per unit of money spent must be equalized across all goods" (assuming it is not a corner solution)

Your Buyer's Problem (\$300 available)

$P_S = 25$
Sweaters \$25

$P_J = 50$
Jeans \$50

Quantity	Total Utility	Marginal Utility	Marginal Benefits per Dollar Spent = (B) / \$25	Quantity	Total Utility	Marginal Utility	Marginal Benefits per Dollar Spent = (D) / \$50
(A)	(B)	(B) / \$25	(C)	(D)	(D) / \$50		
0	0	Blank	Blank	0	0	Blank	Blank
<u>1</u>	100	100	<u>4</u>	<u>1</u>	160	160	<u>3.2</u>
<u>2</u>	185	85	<u>3.4</u>	<u>2</u>	310	150	<u>3</u>
<u>3</u>	260	75	<u>3</u>	<u>3</u>	410	100	<u>2</u>
<u>4</u>	325	65	<u>2.6</u>	4	490	80	1.6
<u>5</u>	385	60	<u>2.4</u>	5	520	30	0.6
<u>6</u>	425	50	<u>2</u>	6	530	10	0.2
7	480	45	1.8	7	533	3	0.06
8	520	40	1.6	8	535	2	0.04

- Your first \$25 should go to 1 sweater, remaining income \$275
- Next \$25 go to second sweater, remaining income is \$250
- Next \$50 go to 1st pair of jeans, remaining income is \$200
- Next \$25 go to 3rd sweater, remaining income is \$175
- Next \$50 go to 2 pair of jeans, remaining income is \$125

Next...

Rem. Income

- Next \$25, 4th sweater, \$100
- Next \$25, 5th sweater, \$75
- Next \$25, 6th sweater, \$50
- Next \$50, 3rd pair of jeans, \$0

I've spent all my
Income, optimal
bundle is

6 sweaters
3 pairs of jeans,



$$\frac{MU_S}{P_S} = \frac{MU_J}{P_J}$$

OR

$$\frac{MU_S}{MU_J} = \frac{P_S}{P_J} = \frac{1}{2}$$

Not done yet, still have money.